ECE 447

Practice Exam

Name: .....

Student ID: .....

Q1 (Maximum Likelihood) The probability mass function (pmf) of a Bernoulli distribution Bern(p) is

$$p_X(x) = \begin{cases} p, & \text{if } x = 1 \\ 1 - p, & \text{if } x = 0 \end{cases}$$

Let $x_1, x_2, \ldots, x_n$ be independent and identically distributed (i.i.d.) samples from the Bernoulli random variable $X \sim \text{Bern}(\theta)$.

a) Explain what maximum likelihood (ML) estimation is doing.

b) Denote $\mathbf{x} = [x_1, x_2, \ldots, x_n]^\top$. Write down the likelihood $L(\theta \mid \mathbf{x})$.

c) Given the n samples, find $\hat{\theta}$, the maximum likelihood estimator (MLE) for θ.

Q2 (Probabilistic ML) Consider the following optimization problem

$$\min_{\boldsymbol{\theta}} \quad \frac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda\|\boldsymbol{\theta}\|_2^2$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n\times m}$, $\boldsymbol{\theta} \in \mathbb{R}^m$.

Discuss how one can derive this optimization objective from a probabilistic perspective on the linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon},$$

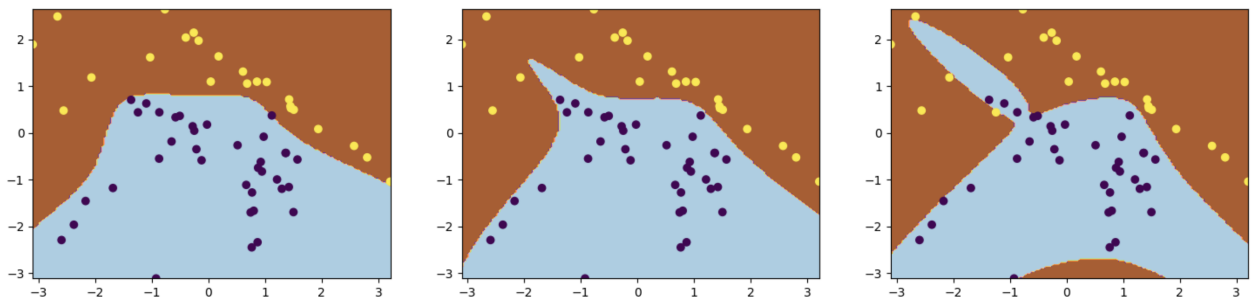
where $\boldsymbol{\epsilon}$ is noise.

Note: The probability density function of the Gaussian distribution $\mathcal{N}(\boldsymbol{m}, \sigma^2\boldsymbol{I})$ is

$$p(\mathbf{y}) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp(\frac{-\|\mathbf{y} - \boldsymbol{m}\|_2^2}{2\sigma^2})$$

Q3 (Classification with Polynomial Logistic Regression) We have used polynomial logistic regression with degree 6 for the classification problem below. The scenarios are:

- Logistic regression
- Logistic regression with ℓ_2 regularization
- Logistic regression with ℓ_1 regularization



Match each figure to the corresponding scenario. Explain your reasoning.

Q4 (Generalization) You have a binary classification problem and a dataset. Your supervisor suggests exploring polynomial features with logistic regression.

Explain the steps you would take to train and evaluate a model, and how you would choose a suitable polynomial degree.

Q5 (Clustering) Consider the following 2-dimensional dataset:

$$\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\1\end{bmatrix}, \begin{bmatrix}6\\5\end{bmatrix}, \begin{bmatrix}7\\5\end{bmatrix}$$

.

We apply K-means clustering with $K = 2$. The initial centroids are:

$$m_1 = \begin{bmatrix}1\\1\end{bmatrix} \qquad m_2 = \begin{bmatrix}7\\5\end{bmatrix}.$$

a) Assignment step. Assign each data point to the nearest centroid using Euclidean distance. Report the cluster sets as

$$C_1 = \{\} \qquad C_2 = \{\}$$

b) Update step. Compute and report the updated centroids after one iteration.

c) Based on the data, would $K = 2$ or $K = 3$ be more appropriate? Explain.

END OF QUESTIONS

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End of Exam